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Batch - 46

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING																		
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026																	
Course Coordinator Name		Dr. Rishabh Mittal																		
Instructor(s) Name		<table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Sujitha Reddy</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M.Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> </table>		Mr. S Naresh Kumar	Ms. B. Swathi	Dr. Sasanko Shekhar Gantayat	Mr. Md Sallauddin	Dr. Mathivanan	Mr. Y Srikanth	Ms. N Shilpa	Dr. Rishabh Mittal (Coordinator)	Dr. R. Prashant Kumar	Mr. Ankushavali MD	Mr. B Viswanath	Ms. Sujitha Reddy	Ms. A. Anitha	Ms. M.Madhuri	Ms. Katherashala Swetha	Ms. Velpula sumalatha	Mr. Bingi Raju
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CourseCode	23CS002PC304	Course Title	AI Assisted Coding																	
Year/Sem	III/II	Regulation	R23																	
Date and Day of Assignment	Week1 – Monday	Time(s)	23CSBTB01 To 23CSBTB52																	
Duration	2 Hours	Applicable to Batches	All batches																	
Assignment Number: 1.3(Present assignment number)/24(Total number of assignments)																				
Q.No.	Question		Expected Time to complete																	

1	Lab 2: Exploring Additional AI Coding Tools beyond Copilot – Gemini (Colab) and Cursor AI Lab Objectives: ☐ To explore and evaluate the functionality of Google Gemini for AI-assisted coding within Google Colab. ☐ To understand and use Cursor AI for code generation, explanation, and refactoring. ☐ To compare outputs and usability between Gemini, GitHub Copilot, and Cursor AI. ☐ To perform code optimization and documentation using AI tools. Lab Outcomes (LOs): After completing this lab, students will be able to: ☐ Generate Python code using Google Gemini in Google Colab. ☐ Analyze the effectiveness of code explanations and suggestions by Gemini. ☐ Set up and use Cursor AI for AI-powered coding assistance. ☐ Evaluate and refactor code using Cursor AI features. ☐ Compare AI tool behavior and code quality across different platforms. Task 1: Statistical Summary for Survey Data ☐ Scenario: You are a data analyst intern working with survey responses stored as numerical lists. ☐ Task: Use Google Gemini in Colab to generate a Python function that reads a list of numbers and calculates the mean, minimum, and maximum values. ☐ Expected Output: ☐ Correct Python function ☐ Output shown in Colab ☐ Screenshot of Gemini prompt and result Question: Write a Python program to check whether a given number is an Armstrong number using user input and clear logic. Code:	Week1 - Monday
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```
def is_armstrong_number(num):  
    num_str = str(num)  
    num_digits = len(num_str)  
    sum_of_powers = sum(int(digit) ** num_digits for digit in num_str)  
    return sum_of_powers == num
```

Output:

```
True  
  
=== Code Execution Successful ===
```

Task 2: Armstrong Number – AI Comparison

- ☐ Scenario:
You are evaluating AI tools for numeric validation logic.
- ☐ Task:
Generate an Armstrong number checker using Gemini and GitHub Copilot.
Compare their outputs, logic style, and clarity.
- ☐ Expected Output:
 - ☐ Side-by-side comparison table
 - ☐ Screenshots of prompts and generated code

Question:

Write a Python program using if-else conditions to check whether a given year is a leap year.

Code:

```
def is_leap_year(year):  
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):  
        return True  
    else:  
        return False  
n= int(input("enter the year: "))  
ans=(is_leap_year(n))  
if ans :  
    print(f"{n} is a leap year")  
else :  
    print(f"{n} is not a leap year")
```


Output:

```
def is_leap_year_optimized(year):  
    return year % 4 == 0 and (year % 100 != 0 or year % 400 == 0)  
  
year = int(input())  
if is_leap_year_optimized(year):  
    print("Leap Year")  
else:  
    print("Not a Leap Year")
```

```
• enter the year: 2020
2020 is a leap year
PS C:\Users\telie\OneDrive\Desktop\AT>
```

Task 3: Leap Year Validation Using Cursor AI

- ☐ Scenario:
You are validating a calendar module for a backend system.
- ☐ Task:
Use Cursor AI to generate a Python program that checks whether a given year is a leap year.
Use at least two different prompts and observe changes in code.
- ☐ Expected Output:
 - ☐ Two versions of code
 - ☐ Sample inputs/outputs
 - ☐ Brief comparison

Question:

Write an optimized Python function that returns True or False to check whether a year is a leap year.

Code:

Output:

```
2016
Leap Year
```

Task 4: Student Logic + AI Refactoring (Odd/Even Sum)

☐ Scenario:

Company policy requires developers to write logic before using AI.

☐ Task:

Write a Python program that calculates the sum of odd and even numbers in a tuple, then refactor it using any AI tool.

☐ Expected Output: ☐ Original code

☐ Refactored code

☐ Explanation of improvements

Question:

Write a Python program to calculate the sum of even and odd numbers in a tuple without advanced functions

Code:

```
def sum_even_odd(numbers):  
    sum_even = sum(num for num in numbers if num % 2 == 0)  
    sum_odd = sum(num for num in numbers if num % 2 != 0)  
    return sum_even, sum_odd  
  
numbers = list(map(int, input("enter the number: ").split()))  
even, odd = sum_even_odd(numbers)  
print(even, odd)
```

```
enter the number: 1 2 5 6 9 87 46 3 51  
54 156
```

```
PS C:\Users\telie\OneDrive\Desktop\AT>
```

Output: